

PaliGemma · arXiv 2407.07726 · Sep 16

Archaeologist

seat 2 · 11+3

SigLIP → PaliGemma → π_0 → OpenPI → BEHAVIOR

Paper club · VLA Architectures · dark academic

Influential priors

What PaliGemma inherits — and how it composes them

PaLI / PaLI-X / PaLI-3 / PaLM-E

Family scaling thesis + transfer-first VLM culture

SigLIP + SoViT

Vision tower: contrastive So400m, shape-optimized

Gemma-2B

Decoder-only LM prior (pretrained, not IT)

CLIP → Flamingo / BLIP-2

Dual-encoder → frozen-vision / Q-Former lineage contrast

Pix2Seq / LocCa / SAM

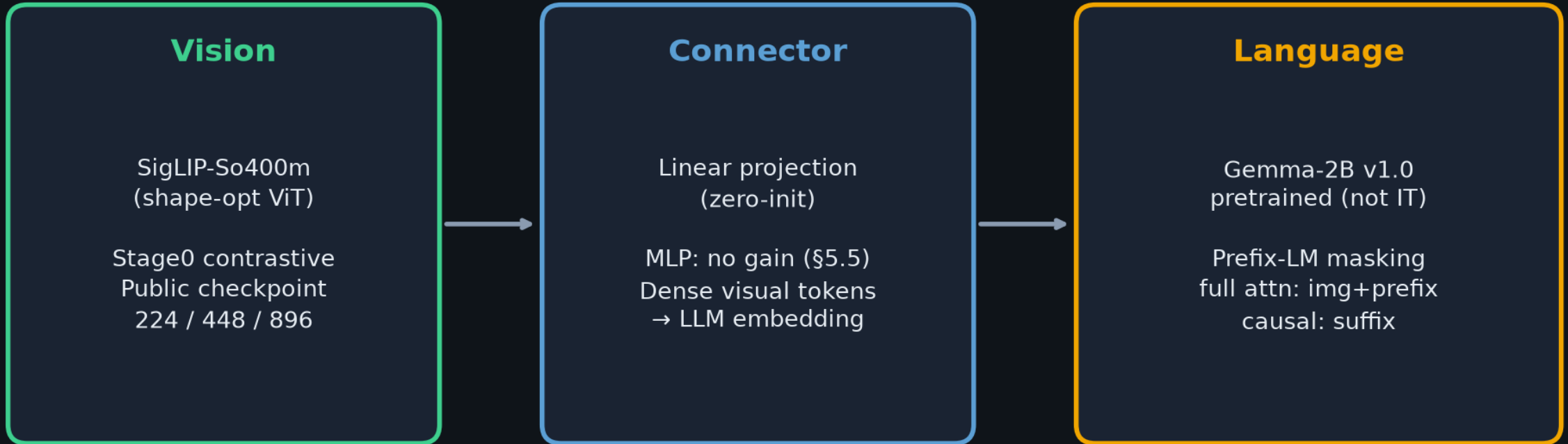
Loc/seg as sequence; grounded caption; mask tokens

Prefix-LM + big_vision

Full attn on prefix; causal suffix; open training stack

Architecture · open base ~3B VLM

SigLIP-So400m + linear connector + Gemma-2B · Prefix-LM



Token layout: [image...] BOS prefix... SEP(\n) suffix... EOS PAD...

loc tokens: 1024 bins · seg tokens: 128 VQVAE codes · skills via task prefixes (no GPT-4 IT)

Perception / grounding backbone — NOT a classical planner or policy

and size. As we will show, this language model is good enough to match or surpass the performance of VLMs using much larger language models, including previous PaLIs. A linear layer projecting SigLIP’s output tokens into the same dimensions as Gemma-2B’s vocab tokens, so they can be concatenated. In early experiments, we found that more complicated alternatives (e.g. MLPs) do not provide a clear advantage, and hence decided to use the simplest option (Sec 5.5).

```
tokens = [image tokens...,
          BOS, prefix tokens..., SEP,
          suffix tokens..., EOS, PAD...]
```

which we place in the front, making image tokens straightforward to interpret without the need for special location markers. The BOS token then marks the start of text tokens. We use `\n` as SEP token, it does not appear in any of our prefixes. We also tokenize SEP separately to avoid it being merged (by the tokenizer) with either the end of the prefix or the beginning of the suffix. In order to maximize model capacity for such a small model, we have full (unmasked) attention on the whole input, *i.e.* the image and prefix tokens. In this way, image tokens can also "lookahead" at the task at hand (prefix) in order to update their representation. The suffix is our output and necessarily covered by an auto-regressive mask, including the PAD tokens. When we mention sequence length (N_{txt}), we typically mean prefix and suffix combined, ignoring image tokens.

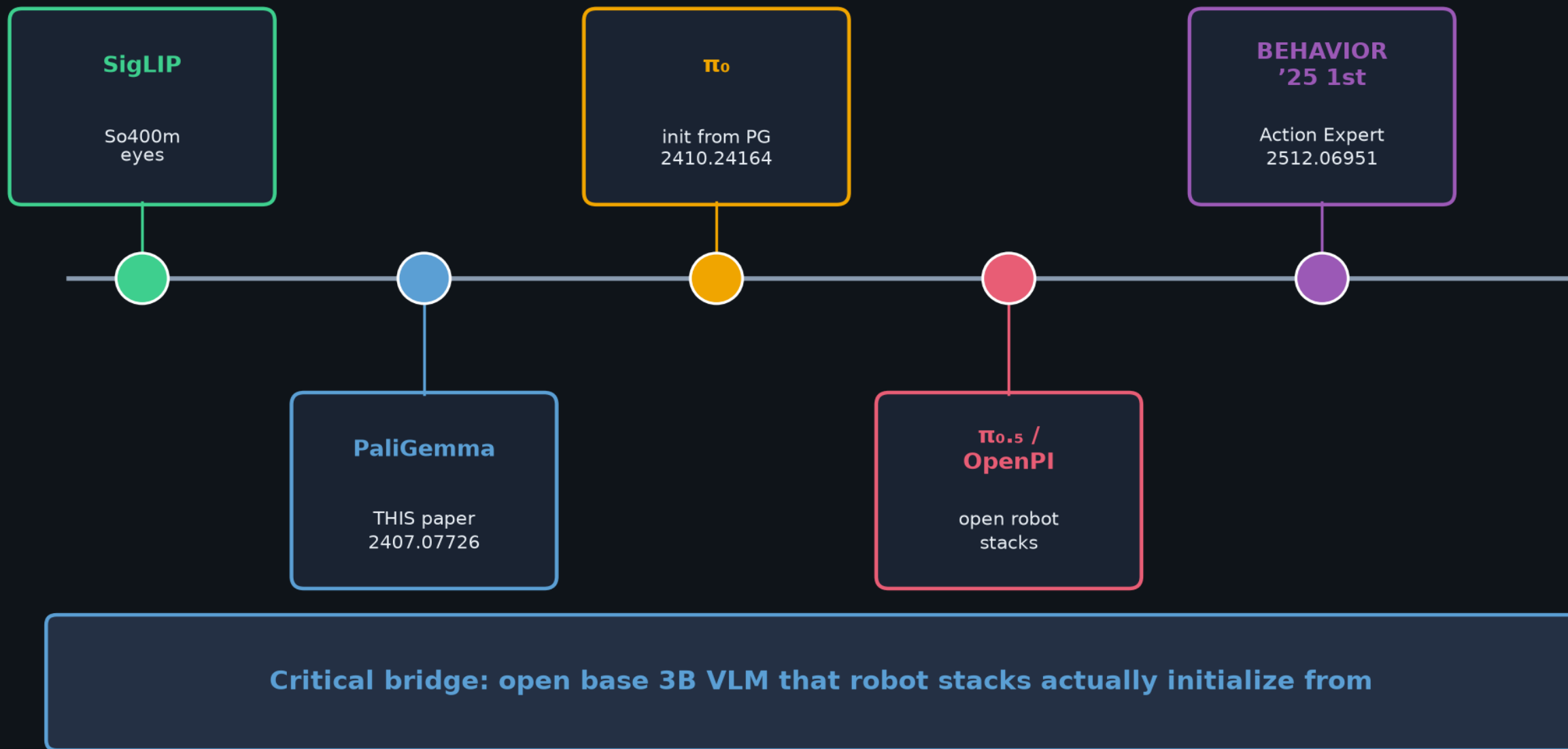
The training of PaliGemma follows the same steps as previous PaLI models, with only small modifications. Training consists of several stages, which we detail in this section:

- ### 3.2.1. Stage0: Unimodal pretraining

Following PaLI-3’s strong experimental results, we use a SigLIP image encoder. While PaLI-3 (and others [6, 26]) use a large image model such as ViT-G, we use the much smaller but similarly strong “shape optimized” ViT-So400m model.

Subsequent timeline

Clear arrow: SigLIP → PaliGemma → π_0 → $\pi_{0.5}$ /OpenPI → BEHAVIOR'25



Club chain · SigLIP → PaliGemma → Action Expert

Why an open 3B base VLM is load-bearing for this club's robot FM work



PaliGemma = the critical open bridge

Closed VLMs cannot seed OpenPI forks. Loc/seg + OCR@448+ = grounding API.
Failures often sit in action expert / stages / data — but these are the eyes.

One-sentence lineage

**PaLI-family + SigLIP/SoViT + Gemma
+ Prefix-LM / LocCa-style tokens
→ PaliGemma → π_0 → $\pi_{0.5}$ / OpenPI
→ BEHAVIOR'25 1st (Action Expert)**

Honest hedges

Private Stage1 mixture → recipe science incomplete even with open weights.
“Uses PaliGemma” ≠ isolating Prefix-LM vs data vs Stage2 resolution.
Downstream robot success is outside this paper’s Table 1.

Club thread · perception vs planner

PaliGemma IS

- ▶ Open base VLM for transfer
- ▶ Loc/seg + OCR grounding API
- ▶ Default π_0 / OpenPI eyes+language
- ▶ Prefix-LM observe+goal shaping

PaliGemma is NOT

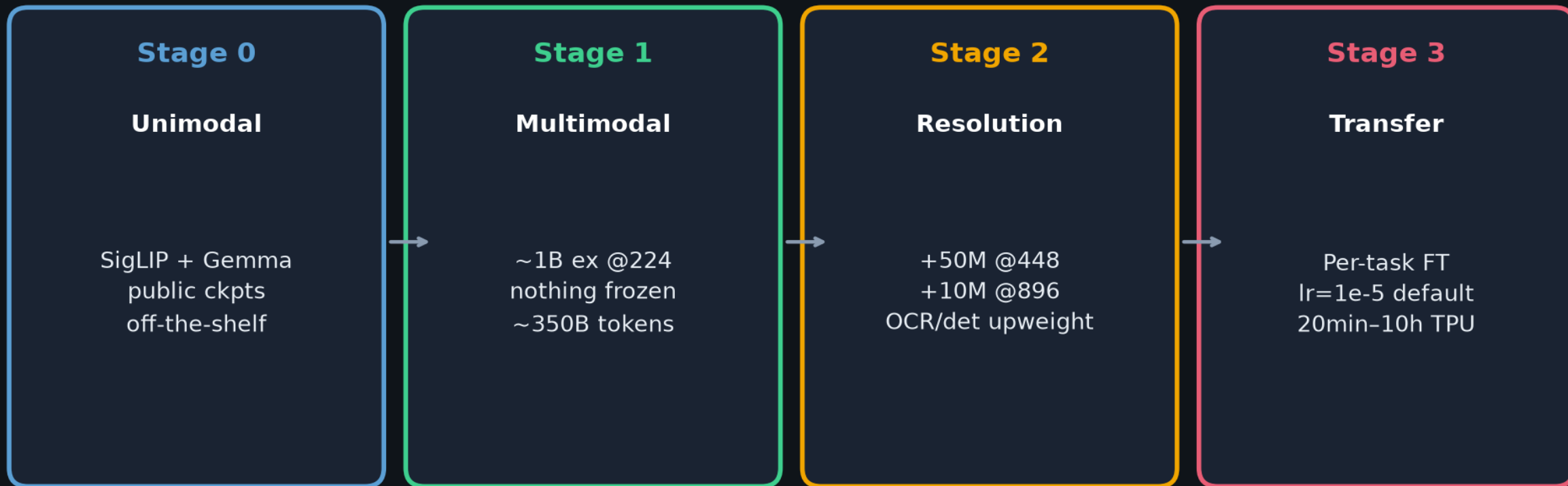
- ▶ A classical motion planner
- ▶ A closed-loop policy
- ▶ An instruction-tuned chat bot
- ▶ Proof of BEHAVIOR wins alone

Hand-offs

Empiricist: LoRA / freeze / res ladder pressure-tests.
Visionary: adaptation (action expert, stages, data) > backbone swap in 2026.

Training stages 0 → 3

Unimodal priors → multimodal skills → resolution → per-task transfer



Key Stage1 choice: do NOT freeze ViT (slow LR warmup) — freezing LM hurts a lot

Prefer released paligemma-pt-{224,448} as substrate for π_0 / OpenPI / BEHAVIOR forks

Archaeologist close

SigLIP → PaliGemma → π_0 → OpenPI → BEHAVIOR · hedges intact